

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. Examination (EC)

November 2012

MA 203: Engineering Mathematics-IV

Date: 03.11.2012, Saturday

Time: 01:30 p.m. to 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION-I

Q-1

(4+3)

(a) Find Fourier Sine and cosine Transform of the function $f(x) = e^{-2x} + 4e^{-3x}$

(b) Find Fourier Transform of the function $f(t) = \begin{cases} te^{-t}, & \text{for } t > 0 \\ 0, & \text{for } t < 0 \end{cases}$

Q-2

(5+5+4)

- (a) Define: Analytic function and orthogonal system. Show that every analytic function $f(z) = u + iv$ defines two families of curves $u(x, y) = 0$ and $v(x, y) = 0$ which forms and orthogonal system.
- (b) In two dimensional fluid flow, the velocity potential is given by $\phi = x^2 - y^2$ find flux function ψ . Is ϕ a Harmonic function?
- (c) Under the transformation $w = z^2$, find the image in w-plane of square at points $(0,0), (2,0), (2,2), (0,2)$ also sketch the image.

OR

Q-2

(5+5+4)

- (a) Find an analytic function in terms of z whose real part is $-r^3 \sin 3\theta$
- (b) Show that the transformation $w = \frac{1}{z}$ transforms all circles and straight lines into the circles and straight lines in w -plane, which circles in z -plane becomes straight line in w -plane and which straight lines are transforms into other straight lines.
- (c) Define bilinear transformation, Find bilinear transformation which maps the points $(0, 1, \infty)$ onto $(-5, -1, 3)$

Q-3

(5+5+4)

(a) Find $L^{-1} \left[\frac{1}{(s+1)(s-2)^2} \right]$ by method of Residues.

(b) Evaluate: $\int_0^{2+i} (\bar{z})^2 dz$ along the line $y = \frac{x}{2}$

(c) State Cauchy integral theorem and evaluate: $\int_{c:|z-1|=1} \frac{3z^2+z}{z^2-1} dz$

OR

Q-3

(5+5+4)

(a) Find the Taylor series expansion for $f(z) = \cos z$ at point $z = \frac{\pi}{4}$

(b) State Cauchy residue theorem and hence evaluate: $\int_{c:|z|=\frac{3}{2}} \frac{4-3z}{z(z-1)(z-2)} dz$

(c) Obtain Laurent's series expansion of function $f(z) = \frac{1}{z^2+4z+3}$ in the regions $|z| < 1$ and $|z| > 3$

SECTION-II

Q-4

(3+4)

(a) Find the Z-transforms of unit impulse function.

(b) Solve the difference equation: $y_{k+1} + \frac{1}{4} y_k = \left(\frac{1}{4}\right)^k$, $k \geq 0$, $y(0) = 0$

Q-5

(5+5+4)

(a) Evaluate $\int_0^1 e^{-x^2} dx$ by the Gauss integration formula for $n = 3$.

(b) Find the interpolating polynomial for (0, 2), (1, 3), (2, 12) and (5, 147) using Langrange's interpolation formula.

(c) Compute $\cosh(0.56)$ using Newton Forward difference formula for the following table

x	0.5	0.6	0.7	0.8
$f(x)$	1.127626	1.185465	1.255169	1.337435

OR

Q-5

(7+7)

(a) Evaluate $\int_0^1 e^{-x^2} dx$ using (i) Trapezoidal rule and (ii) Simpson's $\frac{1}{3^{rd}}$ rule taking $h = 0.2$

(b) Use Newton-Raphson method to solve the transcendental equation: $3x = \cos x + 1$

Q-6

(7+7)

(a) Solve, by Euler's method, the initial value problem $\frac{dy}{dx} = \frac{x-y}{2}$, $y(0) = 1$ over $[0, 3]$,

and using step size $\frac{1}{2}$

(b) Find a real root of the equation $x^3 - 5x - 7 = 0$ using Regula Falsi method.

OR

Q-6

(7+7)

(a) Using Picards method solve the differential equation: $\frac{dy}{dx} = x^2 + 2xy$, $y(0) = 0$.

(b) Apply the fourth order Runge-Kutta method to solve the differential equation:

$\frac{dy}{dx} = x + y$, $y(0) = 1$ for $h = 0.1$
